

CLOUD CERTIFICATION GUIDE

Guide to Prepare for AWS and Azure Certifications

RESOURCE	DETAIL
Category	Cloud Certification & Career Preparation
Coverage	AWS • Azure • DevOps • Security • Containers • IaC • CI/CD
Level	Beginner → Associate → Professional
Format	Structured study guide + exam strategy + practice questions
Language	English

How to use this guide: Learn the concept → map it to a real cloud service → perform a hands-on task → solve scenario questions → review mistakes. Certification preparation should test both knowledge and decision-making.

Important: Certification names, exam objectives, pricing and policies can change. Always verify the current official exam guide before booking an exam.

1. CERTIFICATION ROADMAP

Track	Beginner	Associate / Intermediate	Advanced
AWS	Cloud Practitioner	Solutions Architect – Associate	Solutions Architect – Professional
Azure	AZ-900 Azure Fundamentals	AZ-104 Azure Administrator / AZ-204 Azure Security	AZ-284 Azure Solutions Architect

Recommended for a beginner: Start with one cloud platform instead of studying AWS and Azure equally from day one. Build transferable cloud fundamentals, then learn the second platform through service comparisons.

2. CLOUD COMPUTING FUNDAMENTALS

Concept	What you should know
Cloud computing	On-demand access to compute, storage, networking and managed services.
IaaS	You manage more of the infrastructure; useful for VMs and custom environments.
PaaS	Provider manages more infrastructure so developers focus on applications.
SaaS	Complete application delivered as a service.
Public cloud	Shared provider infrastructure delivered to many customers with logical isolation.
Region	A geographic area containing cloud infrastructure.
Availability Zone	An isolated location within a cloud region, designed for resilience.
Scalability	Ability to handle increased load by adding resources.
Elasticity	Ability to automatically add/remove resources as demand changes.
High availability	Designing systems to remain operational despite component failures.
Fault tolerance	Continuing operation even when a component fails.

Exam habit: When you see words such as highly available, fault tolerant, least privilege, cost optimized or serverless, treat them as architectural clues and evaluate which service characteristics match.

3. AWS CORE SERVICES

Area	Service / Concept	Know this
Compute	EC2	Virtual servers; instance types, AMIs, security groups, scaling and pricing.
Compute	Lambda	Serverless functions; event-driven execution and no server management.
Containers	ECS / EKS	Container orchestration; EKS is managed Kubernetes.
Storage	S3	Object storage; buckets, objects, storage classes, lifecycle, versioning and access.
Storage	EBS	Block storage commonly attached to EC2.
Database	RDS	Managed relational databases with backups, Multi-AZ options and scaling.
Database	DynamoDB	Managed NoSQL key-value/document database.
Networking	VPC	Logical network with subnets, routing, gateways and security controls.
Networking	CloudFront	CDN for caching and delivering content closer to users.
Integration	SQS	Message queue for decoupling producers and consumers.
Integration	SNS	Pub/sub messaging and notifications.
Monitoring	CloudWatch	Metrics, logs, alarms and operational monitoring.
Identity	IAM	Users, roles, policies and least-privilege access.

AWS memory map: EC2 = compute • S3 = objects • EBS = blocks • RDS = relational • DynamoDB = NoSQL • VPC = network
• IAM = identity • CloudWatch = monitoring • SQS/SNS = messaging.

4. AZURE CORE SERVICES

Area	Service / Concept	Know this
Compute	Azure Virtual Machines	Cloud VMs with configurable compute, networking and storage.
Compute	Azure Functions	Serverless event-driven compute.
Containers	AKS	Managed Kubernetes service.
Storage	Blob Storage	Object storage for unstructured data.
Storage	Managed Disks	Block storage for Azure VMs.
Database	Azure SQL Database	Managed relational database service.
Database	Cosmos DB	Globally distributed NoSQL database service.
Networking	Virtual Network	Private networking, subnets, routing and security.
Networking	Azure Front Door	Application delivery and global routing capabilities.
Identity	Microsoft Entra ID	Identity and access management.
Monitoring	Azure Monitor	Metrics, logs, alerts and application monitoring.

Azure memory map: VMs = compute • Functions = serverless • Blob = objects • Azure SQL = relational • Cosmos DB = NoSQL • VNet = network • Entra ID = identity • Azure Monitor = observability.

5. AWS vs AZURE SERVICE MAPPING

Purpose	AWS	Azure
Virtual machines	EC2	Azure Virtual Machines
Serverless functions	Lambda	Azure Functions
Object storage	S3	Blob Storage
Managed relational DB	RDS	Azure SQL Database
NoSQL	DynamoDB	Cosmos DB
Virtual network	VPC	Virtual Network
Kubernetes	EKS	AKS
Identity	IAM	Microsoft Entra ID
Monitoring	CloudWatch	Azure Monitor
CDN / edge delivery	CloudFront	Azure Front Door / CDN offerings

Do not memorize names only. Learn the capability and decision criteria: relational vs NoSQL, synchronous vs asynchronous, stateful vs stateless, managed vs self-managed, and low-latency vs cost-optimized designs.

6. CLOUD SECURITY FUNDAMENTALS

Topic	Core principle
Least privilege	Grant only the permissions required to perform the task.
MFA	Use an additional authentication factor for stronger account protection.
Encryption at rest	Protect stored data using appropriate encryption mechanisms and key management.
Encryption in transit	Use TLS/HTTPS and secure network communication.
Secrets	Do not hard-code credentials in source code; use managed secret/configuration services.
Network segmentation	Separate public-facing resources from private workloads using subnets and routing.
Logging	Collect and retain security-relevant events for detection and investigation.
Backups	Test restore procedures; a backup that cannot be restored is not a reliable recovery.

Common exam clue: If the question asks for secure access without embedding long-lived credentials in an application, look for a role/managed identity or equivalent temporary-credential mechanism.

7. CLOUD NETWORKING

Concept	What to learn
CIDR	Understand IP ranges and how subnet sizes are represented.
Public vs private subnet	Know how routing and gateways determine whether resources can reach the public internet.
Route table	Controls where network traffic is sent.
Internet gateway	Provides a path between a VPC/VNet and the internet when routing and security policies allow.
NAT	Allows private resources to initiate outbound internet connections without making them public.
Load balancer	Distributes traffic across healthy targets.
Security groups / NSGs	Network-level access controls around workloads.
DNS	Maps names to network destinations; understand records and resolution flow.

Practice scenario: A private application server needs to download operating-system updates but must not accept inbound internet traffic. Explain the routing and egress design you would use.

8. DOCKER, KUBERNETES & DEVOPS

Topic	Must know
Docker image	Immutable package containing application code and runtime dependencies.
Container	Running instance of an image with isolated processes/resources.
Registry	Stores and distributes container images.
Kubernetes	Orchestrates containers across a cluster.
Pod	Smallest deployable unit in Kubernetes; one or more tightly coupled containers.
Deployment	Manages replicated application pods and rollout behavior.
Service	Provides stable network access to Kubernetes workloads.
CI	Automatically build/test changes after code changes.
CD	Automate delivery/deployment toward target environments.
Infrastructure as Code	Define infrastructure in version-controlled configuration.

Hands-on task: Containerize a simple Node.js/Java application, publish the image to a registry, deploy it to a managed container platform, expose it through a load balancer and inspect logs.

9. INFRASTRUCTURE AS CODE

Infrastructure as Code (IaC) treats infrastructure configuration as version-controlled, reviewable and repeatable code.

Practice	Benefit
Version control	Review infrastructure changes and roll back when appropriate.
Reusable modules	Avoid repeating the same infrastructure definition.
Plan / preview	Inspect intended changes before applying them.
State management	Track resources managed by the IaC system.
Secrets handling	Keep credentials out of source code and state where possible.
Drift detection	Identify differences between declared and real infrastructure.

Recommended practice project: Define a small VPC/VNet, subnets, a compute service, a database and security rules using IaC. Store the configuration in Git and document the deployment steps.

10. COST OPTIMIZATION

- Right-size compute instead of choosing oversized instances by default.
- Use autoscaling where workloads vary significantly.
- Choose appropriate storage classes and lifecycle policies.
- Delete unused disks, IPs, snapshots and test resources.
- Use reserved/committed pricing only when the workload is predictable and the commitment is justified.
- Prefer managed/serverless services when their operational savings outweigh their price premium.
- Use budgets, alerts and cost dashboards.
- Tag resources so costs can be attributed to teams, projects or environments.

Exam pattern: The cheapest option is not automatically correct. Consider performance, availability, operational effort, traffic patterns and the actual requirement before selecting a cost strategy.

11. HANDS-ON PROJECTS

Project	What to build
Cloud-hosted web app	Deploy a React/Angular frontend, backend API and database with HTTPS and environment variables.
Serverless API	Create API endpoints backed by a serverless function and managed database/storage.
Static website + CDN	Host a static site in object storage and distribute it through a CDN.
Container deployment	Dockerize an application and deploy it to a managed container/Kubernetes service.
CI/CD pipeline	Run tests, build an artifact/container image and deploy automatically to a development environment.
Monitoring project	Add application logs, metrics, health checks, alerts and a simple dashboard.

Placement advantage: A cloud project is much stronger when you can explain architecture, security, cost, deployment, monitoring and failure handling instead of simply saying “I deployed my website.”

12. PRACTICE QUESTIONS WITH ANSWERS

Q1. What is the difference between scalability and elasticity?

Answer: Scalability is the ability to handle increased load by adding resources. Elasticity emphasizes automatically adding and removing resources as demand changes.

Q2. When would you choose object storage instead of block storage?

Answer: Use object storage for files/blobs such as images, videos, backups and documents. Block storage is appropriate when a VM or application needs a mounted block device with filesystem/database semantics.

Q3. Why use a load balancer?

Answer: It distributes requests across healthy targets, improving availability and allowing horizontal scaling.

Q4. What is a security group used for?

Answer: It acts as a network access-control layer around supported resources, allowing or denying traffic according to configured rules.

Q5. Why is IAM least privilege important?

Answer: It limits the damage caused by accidental or compromised credentials by granting only the permissions required for a task.

Q6. What is serverless computing?

Answer: A model where the provider manages the underlying servers and capacity, while the developer focuses on application logic and is commonly billed based on execution/usage.

Q7. Why use a managed database?

Answer: The provider handles many operational tasks such as backups, patching and infrastructure management, allowing teams to focus more on the application.

Q8. What is a VPC/VNet?

Answer: A logically isolated virtual network in the cloud where you define subnets, routes and network security controls.

Q9. What is a CDN?

Answer: A content delivery network caches and serves content from locations closer to users, reducing latency and origin load.

Q10. What is Infrastructure as Code?

Answer: A method of defining infrastructure through version-controlled configuration so environments can be created consistently and reviewed like software.

Q11. What is CI/CD?

Answer: CI automates build/test integration; CD automates delivery or deployment of validated changes toward target environments.

Q12. What is the difference between a container and a virtual machine?

Answer: A VM virtualizes a complete guest operating system; containers share the host kernel while isolating application processes, generally making them lighter to start.

Q13. Why use a queue such as SQS or another message broker?

Answer: To decouple producers and consumers, absorb traffic spikes and support asynchronous processing.

Q14. What is a multi-AZ architecture?

Answer: A design that distributes components across multiple isolated availability zones to reduce the impact of a single-zone failure.

Q15. What is RTO?

Answer: Recovery Time Objective: the target maximum time to restore service after a disruptive event.

Q16. What is RPO?

Answer: Recovery Point Objective: the target maximum amount of data loss measured in time.

Q17. Why should secrets not be stored in Git?

Answer: Repositories can be copied, logged or exposed. Use a secrets-management mechanism and rotate credentials if exposure occurs.

Q18. What is autoscaling?

Answer: Automatically adjusting the number or size of resources based on workload or policy signals.

Q19. What is observability?

Answer: The ability to understand system behavior through telemetry such as logs, metrics and traces.

Q20. How would you reduce cloud costs for an idle development environment?

Answer: Schedule non-production resources to stop when unused, remove orphaned resources, right-size workloads, use appropriate storage classes and monitor spend.

13. EXAM STRATEGY

1. Read the official exam guide and understand the current domains before beginning serious preparation.
2. Build a service map instead of memorizing isolated service names.
3. Practice scenario-based questions because cloud exams often test architectural judgment.
4. Eliminate obviously wrong options first, then compare the remaining choices against the stated requirement.
5. Watch for qualifiers such as most cost-effective, least operational overhead, highly available, secure, real-time and decoupled.
6. Review every practice-test mistake and write why the correct option is correct.
7. Do hands-on labs so service behavior becomes concrete.
8. Do not rely solely on dumps or memorized answer keys; they can be inaccurate and can undermine actual understanding.
9. Simulate the real exam environment with timed practice and no external help.
10. Schedule the exam only when practice performance is stable rather than after one unusually good mock.

14. STUDY RESOURCES

Resource	Use it for
AWS Documentation	Service concepts, architecture, limits, security and configuration details.
AWS Skill Builder	Official learning content and practice resources.
Microsoft Learn	Official Azure learning paths, modules and documentation.
Official exam guides	Current objectives, domains and exam information.
Practice-test platforms	Timed question practice and identifying weak domains.
YouTube / courses	Concept explanations and guided labs; verify technical details against official docs
Hands-on cloud labs	Building and troubleshooting real architectures.

Resource rule: Use third-party courses and practice tests for explanation and repetition, but use the official provider documentation/exam guide as the final authority for current certification requirements.

15. 4-MONTH STUDY SCHEDULE

Month	Focus	Suggested output
Month 1	Cloud fundamentals + core compute/storage/networking	On-premise web app + service notes
Month 2	Databases, IAM, monitoring, serverless and architecture	Two on-labs + 2 timed practice sets/week
Month 3	Containers, Kubernetes, IaC, CI/CD and security	Docker deployment + IaC project + pipeline
Month 4	Practice tests, weak domains and revision	Full mocks + error log + final revision

Weekly routine: 5 days concept + lab work • 1 day timed practice • 1 day revision and error analysis.

16. FINAL CERTIFICATION CHECKLIST

- ✓ Read the current official exam guide.
- ✓ Know the major services and their primary use cases.
- ✓ Understand IAM/security and least-privilege concepts.
- ✓ Understand networking fundamentals.
- ✓ Know storage and database trade-offs.
- ✓ Understand high availability and disaster recovery basics.
- ✓ Practice cost-optimization scenarios.
- ✓ Build at least one real cloud project.
- ✓ Use containers and understand basic Kubernetes concepts.
- ✓ Understand CI/CD and Infrastructure as Code.
- ✓ Complete timed practice tests.
- ✓ Maintain an error notebook.
- ✓ Revisit every weak domain.
- ✓ Verify current exam registration and policy information before booking.
- ✓ Do not depend on unauthorized exam dumps.

Final principle: Certifications can validate knowledge, but hands-on projects demonstrate that you can apply it. For placement preparation, combine the certification with a deployable project, GitHub documentation and the ability to explain your architecture.